MILITARY SPECIFICATION

SIGHT, REFLEX: XM70E1

1. SCOPE

1.1 This specification covers one type of reflex sight equipped with a parallelogram linkage to elevate the line of sight in synchronism with a weapon, a height adjustment for the observer, and a range knob for super-elevation angles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10547304	Sight, Reflex: XM70E1

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PACKAGING DATA SHEETS

10547304 Packaging of Sight, Reflex: XM70E1

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.).

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The sight shall be fabricated in accordance with Drawing F10547304 and drawings pertaining thereto.

3.3 General specifications.- The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-O-13830 and MIL-F-13926.

3.3.1 MIL-O-13830.

(a) Lenses
(b) Cleanliness

3.3.2 MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Storage at extreme temperature.- The sight shall not evidence damage after storage at -65°Fahrenheit (F) and +160°F.

3.5 Operability at extreme temperature.- Requirements 3.5.1 to 3.5.4 shall be met while the sight is thermally stabilized at -40°F and at +150°F.

3.5.1 Light intensity control and light filament selector switch.- The light intensity control and light filament selector switch shall be manually operable.

3.5.2 Arm 10547473.- The torque values exhibited by Arm 10547473 shall not exceed the values demonstrated for each position specified in 3.7.14 by more than 10 pound-inches.

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3.5.3 Elevation mechanism and height adjustment control.- The elevation mechanism and height adjustment control shall be manually operable throughout their respective excursions.

3.5.4 Range knob.- The torque required to rotate the range knob, out of each detent position, through the range specified in 3.7.11 (Table I), shall not exceed the value exhibited when meeting 3.7.15 by more than 10 pound-inches.

3.6 Humidity.- The sight shall not evidence damage after exposure to a 97% relative humidity (RH) for 24 hours at +160°F followed by four hours at between +60° and +90°F with condensation taking place on the sight.

3.7 Performance between +60°F and +90°F.- Except where otherwise specified, the sight shall meet the requirements of 3.7.1 to 3.7.12 after orientations "a" to "e" inclusive have been effected. (Reference Figure 1 and Drawing F10547304).

(a) The angular difference between line "x" and a perpendicular to line "w" shall be 17° 18' 24" ± 24" for 3.7.2 through 3.7.12 and 17° ± 1° for 3.7.1.

(b) Line "w" shall be horizontal.

(c) Range Knob "250" on the white scale shall be in coincidence with Housing 10547322 index.

(d) Arm 10547473 shall be secured in its operating position.

(e) Applied direct current (DC) shall be of an appropriate value and polarity.

3.7.1 Vibration.- Throughout the entire vibration, the range knob shall not move out of range detent. The sight shall evidence no damage or electrical circuit malfunction during or after the vibration specified herein, with Bell Crank 10547317 immobilized, electrical potential applied, and Arm 10547473 in an operating and stowed position. The initial optical-mechanical alignment demonstrated while meeting 3.7.2 shall not have changed more than one mil subsequent to either vibration. There shall be no flickering of the projected reticle image during vibration.

3.7.1.1 Vibration "A".- This vibration shall be for two hours, in each plane of three mutually perpendicular planes, at the sinusoidal frequencies and amplitudes depicted in Figure 2. During each first hour period, the frequency shall be cycled logarithmically from 5 to 500 to 5 cps for four cycles each of 15 ± 1 minute duration. A fifteen minute dwell shall occur at each of the most significant resonant frequencies (not exceeding 4) exhibited by the sight during the first hour of each two hour vibration in each plane. Arm 10547473 shall be sequentially located in a stowed and operating position when meeting this requirement, and Reticle Lamp 10547031 shall not be subjected to the vibration dwells specified herein.

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3.7.1.2 Vibration "B".- The sight's projected reticle image shall not be displaced more than ±5 mils during this vibration from its initial position prior to vibration. This vibration shall be in a vertical plane for 5 minutes ± 15 seconds at a frequency of 32 cps with an amplitude of 0.015 inch (0.03 inch total excursion).

3.7.2 Optical-mechanical alignment.

3.7.2.1 Azimuth alignment.- The line of sight through the geometric axis of the projected reticle pattern shall be adjustable and capable of being locked at any angle between 2.5° right and 0.5° left of a plane that is parallel to the vertical plane through reference line X (See Figure 1).

3.7.2.2 Elevation alignment.- The line of sight through the geometric axis of the projected reticle pattern shall be parallel to reference line W (Figure 1) within 4.0 mils with Arm 10547473 in the operating position and Bell Crank 10547317 mobilized.

3.7.3 Optical-mechanical alignment retention.- The optical-mechanical alignment demonstrated by the sight in 3.7.2 shall not change more than ±0.5 mil when Arm 10547473 is repeatedly moved from its operating to stowed to operating position, or when the sight is repeatedly elevated 177 mils and depressed 408 mils or removed and reinserted into its orientation medium.

3.7.4 Extraneous light emission.- The light source shall not be visible except as observed through the lenses.

3.7.5 Projected reticle image definition.- The projected reticle image shall be clearly defined with no visible breaks when viewed from a normal operating position against a bright background (such as the reflection of a clear sky from water) or when viewed in a darkened area.

3.7.6 Projected reticle image pattern accuracy.- The angle subtended by the inner circle of the projected reticle image pattern shall be 30 ± 1.0 mil.

3.7.7 Parallax.- When viewing at a distance of eight to ten inches from the beamsplitter, parallax between the reticle image center and the target shall not exceed 2.0 mils over the full aperture of the beamsplitter when viewing a target at a range of 1000 ± 200 meters.

3.7.8 Image tilt.- The vertical line of the projected reticle image shall be plumb within 1.0 degree when meeting 3.7.2.

3.7.9 Elevation accuracy.- With the elevation position established in 3.7.2.2, the angular changes in the line of sight between 214 mils depression and 177 mils elevation shall be within 2.0 mils of the same angular changes in Refernce Line "W". The angular changes in the line of sight, between 214 mils depression and 408 mils depression shall be within 4.0 mils of the same angular change in Reference Line "W".

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3.7.10 Plumb travel.- The line of sight established in 3.7.2 shall track a plumb line within 2.0 mils from 177 mils elevation to 213 mils depression, and shall track a plumb line within 4.0 mils from 214 mils depression to 408 mils depression.

3.7.11 Range control accuracy.- Rotation of the range knob from zero orientation shall cause an angular change in the line of sight within the specified tolerance as indicated in Table I.

TABLE I

RANGE (Meters)	SIGHT CHANGE (Mils)	TOLERANCE (Mils ±)
250 (White)	0	
500 (White)	12.01	1.5
750 (White)	18.24	2.0
1000 (White)	28.08	2.0
250 (Yellow)	-8.55	2.0
500 (Yellow)	34.90	3.0
750 (Yellow)	71.18	3.0
1000 (Yellow)	111.37	3.0
1250 (Yellow)	156.10	3.0
1500 (Yellow)	212.32	2.5

3.7.12 Height adjustment mechanism.- Movement of the height adjustment mechanism throughout its entire excursion shall not cause the line of sight to deviate in excess of 2.0 mils from a plumb line.

3.7.13 Electrical controls.- Rotation of the light intensity control, from stop to stop in a clockwise (cw) and counterclockwise (ccw) direction, shall gradually vary reticle lamp light intensity in both positions of the filament selector switch. Full cw to full ccw rotation of this control shall produce a gradual and regular dimming of the reticle, with no breaks in continuity. With the control in an extreme ccw position, filaments of the dual filament reticle lamp shall emit no light.

3.7.14 Arm 10547473 operability.- When applied at the stop on Housing 10547482 casting, a torque of between 20 and 30 pound-inches shall move Arm 10547473 0.002 inch from its operating position; a torque of between 45 and 60 pound-inches shall move it from an operating to stowed position; and a torque of between 30 and 45 pound-inches shall move it out of a stowed position.

3.7.15 Range knob torque.- The torque required to rotate the knob, out of each detent position, through the range specified in 3.7.11 (Table I) shall be between three and ten pound-inches. The knob shall positively index in each detent position.

3.7.16 Sealing.- Housing 10547482 shall evidence no leakage in excess of 0.2 pounds-per-square-inch, gage pressure (psig) when subjected to an internal pressure of 6 ± 1.0 psig for at least two hours.

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3.7.17 Interior atmosphere.- The interior atmosphere of Housing 10547482 shall have a dew point of -25°F, or lower, with an atmospheric pressure of 0.10 to 0.25 psig.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-O-13830 and MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) sights plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 and MIL-O-13830 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II and III shall constitute the minimum inspection to be performed by the supplier after first article approval, compliance with certification requirements as specified in 4.4.3, and prior to Government acceptance or rejection by item or lot.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Vibration (General and Vibration B	3.7.1, 3.7.1.2	4.6.4.1, 4.6.4.1.2
102. Optical-mechanical alignment; azimuth and elevation	3.7.2.1, 3.7.2.2	4.6.4.5.1, 4.6.4.5.2
103. Optical-mechanical alignment retention	3.7.3	4.6.4.6
104. Elevation accuracy	3.7.9	4.6.4.7
105. Plumb travel	3.7.10	4.6.4.8
106. Range control accuracy	3.7.11	4.6.4.9
107. Height adjustment mechanism	3.7.12	4.6.4.10
108. Extraneous light emission	3.7.4	4.6.4.11
109. Projected reticle image definition	3.7.5	4.6.4.12
110. Projected reticle pattern accuracy	3.7.6	4.6.4.13
111. Parallax	3.7.7	4.6.4.14
112. Image tilt	3.7.8	4.6.4.15
113. Electrical controls	3.7.13	4.6.4.16
114. Arm 10547473 operability	3.7.14	4.6.4.17
115. Range knob torque	3.7.15	4.6.4.18
116. Sealing	3.7.16	4.6.4.19
117. Interior atmosphere	3.7.17	4.6.4.20
118. Cleanliness	3.3.1(b)	4.6.4.21
MINOR: NONE DEFINED		

NOTE: The tests in Table II shall be conducted in accordance with the test condition requirements in the "Test Facilities" requirement of MIL-F-13926.

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Control tests.

4.4.1 Special sampling.- Samples shall be selected from lots at random using inspection level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. The samples shall have met all the requirements and tests outlined in Table II prior to and after being subjected to the following examinations and tests.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Storage at extreme temperatures	3.4	4.6.1
302.	Operability at extreme temperatures	3.5	4.6.2
303.	Light intensity control and light filament selector switch	3.5, 3.5.1	4.6.2.1
304.	Elevation mechanism and height adjustment control	3.5, 3.5.3	4.6.2.2
305.	Range knob	3.5, 3.5.4	4.6.2.3
306.	Arm 10547473	3.5, 3.5.2	4.6.2.4

4.4.1.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.4.2 First article sample.- Unless otherwise specified in the contract, test(s) listed herein shall be conducted on one unit of the initial production sample and shall not again be performed on the remaining production quantities, provided the manufacturing processes and techniques used to produce the samples have not been changed or revised.

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Humidity	3.6	4.6.3
304.	Vibration (General) and Vibration A	3.7.1, 3.7.1.1	4.6.4.1, 4.6.4.1.1

4.4.3 Certification.- Certification shall include test data and results for all characteristics specified in Table IV. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract

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production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

TABLE IV

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Materials	3.1	Applicable drawings
Fabrication	3.2	Standards, and
General specification (MIL-F-13926)	3.3.2	Specifications
General specification (MIL-O-13830)	3.3.1(a)	

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provided for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (IEL) when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.- Inspection equipment provided in accordance with 4.5.2.2 shall meet the following requirements:

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a. Environmental and mechanical test equipment.- Temperature, humidity, and vibration test equipment shall conform to applicable requirements in the "Test Facilities" requirement of MIL-F-13926. Standard vibration equipment provided shall have provisions for controlling the frequency and amplitude of vibration in accordance with 3.7.1.1 and 3.7.1.2. Vibration equipment shall have provisions for holding the sight securely throughout vibration tests, and allow its orientation in three mutually perpendicular planes for Vibration "A" and in one plane only for Vibration "B" to simulate the projector's position when mounted for use. A standard time measuring instrument shall be provided to control vibration periods. A multi-station pressure tester having at least the functional accuracies of Item 1 in 4.5.2.4 shall be provided along with a dew-point tester and regulator. A spring scale and gage (Items 5 and 8 in 4.5.2.4) to measure force values shall also be provided. Unless otherwise specified, test equipment herein shall be of standard commercial quality.

b. Optical test equipment.- Equipment provided shall include a means to cross level the sight and a medium to securely hold the sight while accurately positioning it along specified optical and geometric lines. It shall also accurately provide specified excursions in azimuth, elevation, and depression (see Items 6 and 7 in 4.5.2.4). A ten to fifteen power (10X to 15X) auxiliary telescope having a field of view (FOV) of at least 30 mils and a reticle with acceptance limits for pertinent optical requirements shall be provided. A collimating projector with a target reticle having acceptance or rejection markings shall be provided. Also, to facilitate inspection during vibration, a target circle shall be provided having a diameter equal to the length of a chord of an arc which subtends a ten mil angle.

c. Electrical test equipment.- An electrical test circuit external to the sight shall be provided. When used in conjunction with an electrical power source, it shall provide the various direct current (dc) potentials in proper polarity required by individual tests. It shall not be subject to the mechanical vibration of the vibration tests, and it shall be connected electrically to items under test by flexible conductors. This test circuit shall incorporate the required electrical instrumentation (see 4.5.2.4, Items 3 and 4) to measure and monitor specified electrical characteristics of the items under test.

4.5.2.4 Test equipment accuracy.- In conjunction with 4.5.2.3, test equipment shall have at least the accuracy depicted for the following equipment:

ITEM	NOMENCLATURE	DESCRIPTION
1.	Pressure tester - two station	Depicted on Drawing F8565556
2.	Dew-point tester w/regulator	Standard commercial quality
3.	Voltmeter	0-50 vdc; full scale accuracy ±2%
4.	Ammeter	0-1.0 Amp dc; full scale accuracy ±2%

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5.	Spring scale	Standard commercial quality
6.	Inspection Aid (Gunner's Quadrant)	Depicted on Drawing A8658940
7.	Fixture, Final Inspection	Depicted on Drawing F10547466
8.	Force gage	0.002 inch thick with attached weight of 2 to 4 grams

4.6 Test methods and procedures.

4.6.1 Storage at extreme temperatures.- Utilizing test equipment conforming to 4.5, gradually change chamber temperature to prevent thermal shock until the external surface of the sight is thermally stabilized at each extreme temperature specified in 3.4. Maintain each extreme temperature for an uninterrupted period of at least 24 hours. Upon completion of each exposure the temperature shall be gradually returned to between +60°F and +90°F. Subsequent to completion of each temperature excursion, the sight shall be examined to determine that no evidence of damage exists.

4.6.2 Operability at extreme temperatures.- Utilizing test equipment conforming to 4.5, gradually change chamber temperatur to prevent thermal shock until the external surface of the sight is thermally stabilized at each extreme temperature specified in 3.5. Maintain each extreme temperature for only the uninterrupted period of time required to perform the tests and examinations specified herein. The tests and examinations in 4.6.2.1 to 4.6.2.4 shall be performed to verify manual operability of mechanical subassemblies at each extreme temperature in compliance with 3.5. Upon completion of all examinations at each extreme temperature, the temperature shall be gradually returned to between +60°F and +90°F.

4.6.2.1 Light intensity control and light switch.- Manually rotate the light intensity control knob throughout its range, in a clockwise (cw) and counterclockwise (ccw) direction. Determine that no binding is evident using normal hand pressure and that the control knob is secure on its shaft. The light switch toggle when manually operated shall be examined for ease of manipulation with positive indexing in both "ON" positions.

4.6.2.2 Elevation mechanism and height adjustment control.- With the height adjustment knob tightened (cw) rotate the sight assembly vertically around its normal pivot shaft to assure that the bushings have not seized on the shaft. When conducting this test, it shall be observed that the height adjustment knob secures the height setting and that the sight is free to rotate unless the bellcrank is also secured. The height adjustment control knob shall now be rotated ccw while observing release of the bearing rod and free movement of the elevation mechanism. Manually rotate the height adjustment control knob cw and determine that the height adjustment mechanism will secure at any desired point throughout its excursion range.

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4.6.2.3 Range control knob.- Utilize the test procedure in 4.6.4.18, and determine that range knob torque measurements do not exceed the values measured in 4.6.4.18 by more than 10 pound-inches at each extreme temperature in 3.5.

4.6.2.4 Arm 10547473.- Utilize the test procedure in 4.6.4.17, and determine that Arm 10547473 torque requirements do not exceed the values obtained in 4.6.4.17 by more than 10 pound-inches at each extreme temperature in 3.5. Operating and stowed positions shall be as specified on Drawing F10547304. Observe that Arm 10547473 moves freely between operating and stowed positions and determine that the mechanism provided will fasten it in both positions.

4.6.3 Humidity.- This test shall be performed subsequent to the sealing test and prior to the interior atmosphere test. Except as specified herein, conduct this test in accordance with the "Test Facilities" and "Humidity (Procedure III)" requirements of MIL-F-13926. The sight shall be exposed for at least 24 hours to a 97% relative humidity (RH) at an ambient temperature of +160°F. Chamber temperature shall then be lowered to between 60° and +90°F with condensation and the sight exposed to this condition for at least four hours - within which time chamber temperature may be adjusted to obtain other condensation periods. The humidity phase of this test shall now cease and the external surface of the sight shall be dried by wiping all accessible areas. Drying by air blast or in an oven shall not be permitted. Pins of electrical connectors that are not exposed when in use shall be covered during this test.

4.6.4 Performance between +60° and +90°F.- Unless otherwise specified, the following tests shall be performed using, as applicable, the test set-up and test procedure in subparagraphs "a" and "b" respectively, and operating instructions associated with final inspection fixtures.

a. Test set-up.- On a cross leveled surface position and secure the medium (fixture) provided to hold and position the sight. Locate on the fixture axes which correspond to axes "X" and "W" on the sight (see Figure 1). Axis "X" is in a plumb longitudinal plane, and axis "W" is in a level longitudinal plane. Establish in space, a geometric axis parallel to the planes containing axes "X" and "W". Vertical and horizontal lines of targets and reticles to be used in the following tests shall be made plumb and level, respectively. The electrical dc power source, dc voltammeter, and time measuring instrument shall be conceniently located relative to the fixture.

b. Test procedure.- Securely position the sight in its fixture in accordance with axes on the sight designated "X" and "W" in Figure 1. Make the necessary electrical connections to apply and verify a sight input potential of 28 vdc with provision for monitoring sight input current during tests. With the filament selector switch in an "ON" position, adjust the sight's light intensity control for optimum viewing of the projected reticle image. While maintaining the parallelism of the geometric

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axis established in the test set-up, the geometric axis shall now be made coincident with the cross-over point of both target and reticle crosshair patterns and the intersecting point of the sight's projected reticle crosshair pattern image - appearing in the central area of the beamsplitter, with Arm 10547473 in an operating position. This established geometric axis shall be the optical line of sight in the following tests and examinations.

4.6.4.1 Vibration.- Prior to vibration, perform test 4.6.4.5, determine compliance with 3.7.2, and record the optical-mechanical alignment demonstrated by the sight. Utilize test equipment conforming to 4.5 and apply 28 vdc to the sight's input in proper polarity. Throughout vibration, the sight shall be under constant surveillance to detect any evidence of optical, mechanical, or electrical component failure or malfunction, and that there is no movement out of detent of the range knob from any preselected position during vibration. Except during dwells in 4.6.4.1.1, electrical circuits shall be monitored to verify that no input current change occurs - indicative of intermittent electrical connection, electrical short circuit, or component malfunction in either "ON" position of the filament selector switch. The constant current drawn by each filament of the dual filament lamp - or between lamps - need not be of identical value. Subsequent to vibration, the sight shall be examined for evidence of damage and again subjected to test 4.6.4.5 to determine compliance with 3.7.1.

4.6.4.1.1 Vibration "A".- Perform this test in compliance with requirements of 3.7.1 and 3.7.1.1 and in accordance with Procedure I (Parts 1, 2 and 3) of MIL-STD-810 and the specific procedure herein and in 4.6.4.1. During this test, set the light intensity control for maximum reticle lamp illumination. The range knob shall be placed in a different detent position and the filament selector switch shall be positioned in an alternate "ON" position, each time Arm 10547473 is relocated during each first hour vibration in each plane. During the first hour of each two hour vibration in each plane, the arm (Arm 10547473) shall be in a stowed position for one half hour and in an operating position for the remaining half hour. Record throughout the first hour of vibration in each plane all predominant resonant frequencies exhibited by the sight and note the position of the arm at that time. Upon completion of the first hour of vibration in each plane, select the four most prominent resonant frequencies of all those recorded regardless of the position of the arm. Remove the reticle lamp and during the remaining one hour vibration in each plane, vibrate the sight for not less than 15 minutes at each one of the four resonant frequencies previously selected with the arm in the position noted when the resonant frequency was first recorded. Upon completion of the entire vibration test (of two hours in each plane for a total of six hours), the sight shall be examined as specified in 4.6.4.5 with reticle lamp 10547031 reinserted.

4.6.4.1.2 Vibration "B".- Perform this test in compliance with requirements of 3.7.1 and 3.7.1.2, procedures herein and in 4.6.4.1 with the light intensity set for optimum viewing of the projected reticle image.

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The range knob shall be set with 250 on the white scale aligned with Housing 10547322 index. Prior to vibration, the geometric center of the projected reticle image shall be made coincident to the intersection of a crosshair and this intersection shall be located at the center of a reticle displacement target circle. The filament selector switch shall be in one "ON" position for half the vibration period and in an alternate "ON" position for the remaining half. Throughout vibration, the observer shall locate his eye eight to ten inches from the beamsplitter in the direction of the projector lamp housing to verify that no flicker occurs in the projected reticle image and that the coincidence established herein remains within the circumference of the reticle target circle. Upon completion of the entire vibration test, the subsequent to vibration examination specified in 4.6.4.5 shall be performed.

4.6.4.5 Optical-mechanical alignment.- This test shall employ a medium to hold and position the sight in compliance with 3.7. Prior to inserting the sight into the medium provided, utilize an auxiliary telescope whose line of sight has been orientated in accordance with 3.7.2 and when its geometric axis is in a vertical longitudinal plane that is parallel to the vertical longitudinal plane containing Line "X" (Figure 1) subsequent to sight insertion. Record the results obtained in 4.6.4.5.1 and 4.6.4.5.2.

4.6.4.5.1 Azimuth alignment.- After mounting the sight, observe that its line of sight is in compliance with 3.7.2.1 and passes through the central area of the beamsplitter. Record the results of this examination. Adjust the azimuth positioning screw over its full range of travel and observe that the line of sight through the geometric axis of the projected reticle pattern can be adjusted through the range specified in 3.7.2.1.

4.6.4.5.2 Elevation alignment.- Observe that the line of sight through the geometric axis of the projected reticle image is in coincidence with its respective horizontal target line, with in the tolerance specified in 3.7.2.2. This check shall be made with line "W" in a level plane.

NOTE: Subsequent to vibration this examination shall be again repeated to determine compliance with the line of sight retention requirement of 3.7.1.

4.6.4.6 Optical-mechanical alignment retention.- This test consists of repeating 4.6.4.5: (a) subsequent to vibration tests in 4.6.4.1; (b) upon elevating and depressing the sight (in accordance with the procedure specified herein); and (c) after completion of each one of each three cycles of the handling cycles specified herein. The recorded data in 4.6.4.5 shall be utilized to verify compliance with 3.7.1 and 3.7.3. Elevation and depression of the sight shall be in a vertical longitudinal plane 177 mils above line "W" (Figure 1) and then 408 mils below line "W". Prior to and after completing each operational cycle, as defined herein, verify that a line of sight in the established geometric axis (established in 4.6.4.5) intersects the center of the projected reticle image in elevation and depression

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within specified limits. Each handling cycle shall consist of moving Arm 10547473 from its operating to stowed to operating positions; elevating and depressing the sight as previously specified in this test; and removing the sight from its holding medium and again reinserting and repositioning the sight in the holding medium.

4.6.4.7 Elevation accuracy.- In accordance with final inspection fixture operating instructions, secure the sight at the elevation station and at the stations in depression specified. Observe that the line of sight through the geometric axis of the projected reticle image is in coincidence with its respective horizontal target line, within the tolerance specified in 3.7.9. This check shall be made at approximately 177 mils elevation, 0 elevation, 213 mils depression, and 408 mils depression.

4.6.4.8 Plumb travel.- In accordance with final inspection fixture operating instructions release and permit the elevation mechanism to travel the excursion in elevation and depression specified. Observe that the line of sight through the geometric axis of the projected reticle image is in coincidence with the plumb target line within the tolerance specified in 3.7.10.

4.6.4.9 Range control knob accuracy.- Rotate the range knob to each range position specified in Table I. To determine compliance with 3.7.11, observe that the resultant angular change in the line of sight is within the tolerance indicated for each induced sight change in accordance with Table I. Observe that each detent exhibits positive indexing. Observe that at each detent position the corresponding knob scale figure is indicated by the index pointer.

4.6.4.10 Height adjustment mechanism.- Utilize a target and adjust the sight's height throughout the adjustment range. To determine compliance with 3.7.12, observe that there is no deviation in the line of sight greater that the amount specified, when the sight is periodically clamped and sighted throughout the entire excursion. This observation shall be made at points near the upper and lower travel limits and at two points approximately 1/3 of travel distance from the top and bottom respectively.

4.6.4.11 Extraneous light emission.- Energize the projector with 28 vdc and adjust the light intensity control to obtain maximum light output from the reticle lamp in a darkened area (with not more than 0.01 foot candle of illumination). Perform a visual examination of the projector to determine that no light is emitted from the projector except through the lenses as defined by the limits prescribed by the reticle drawing (D10543371).

4.6.4.12 Projected reticle pattern definition.- In a darkened area - as specified in 4.6.4.11 - adjust the light intensity control to obtain optimum viewing of the projected reticle pattern image, and conduct an examination to determine that its pattern is without breaks and clearly defined. After a lapse of 15 minutes - a light adaption period - employ

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a uniformly illuminated field of view of at least 2300 foot lamberts. The light intensity control shall now be adjusted to determine that a position exists where the projected reticle pattern image is clearly visible.

4.6.4.13 Projected reticle pattern accuracy.- Adjust the light intensity control for optimum viewing and establish coincidence between the intersection of the horizontal and vertical lines on the target and those of the projector's reticle image. Examine the entire circumference of the projector's 30 mil circle to determine that it is within the two concentric circles which subtend angles of 29 and 31 mils respectively.

4.6.4.14 Parallax.- The observer shall locate his eye eight to ten inches from the beamsplitter in the direction of the sight housing along the line of sight depicted by Drawing F10547304. Adjust light intensity control knob for optimum viewing. Align both the vertical and horizontal reticle line of the sight close to but not touching the vertical and horizontal lines on targets, located at an actual or simulated distance of 1000 meters. The observer shall slightly move his head up and down and from right to left, while observing both target lines and both reticle lines. The observer shall discover no line displacement in excess of the limit specified over the full aperture of the beamsplitter.

4.6.4.15 Image tilt.- Perform this test with a target line that is plumb, aligned with the vertical longitudinal plane containing the line of sight established in 4.6.4.5 and with one extremity of the projector's reticle image vertical line in coincidence with the target's plumb line - at the bottom of the 31 mil circle. The other extremity of the projector's reticle image vertical line shall now be observed at the top of the 31 mil circle to verify that it is not displaced more than 0.4 mil from the plumb target line.

4.6.4.16 Electrical controls.- Apply 28 vdc to the projector, position filament selector switch sequentially in each "ON" position and rotate the light intensity control knob from its extreme ccw position of minimum lamp illumination to the extreme cw position of maximum lamp illumination. Determine that a progressive increase in lamp emitted light occurs with cw rotation of the control and a progressive decrease with ccw rotation - in both positions of the filament selector switch. With the control in an extreme ccw position and the switch sequentially placed in alternate "ON" positions, it shall be observed in a darkened area (having not more than 0.01 foot candle of illumination) that no light is emitted from the lamp. Observe compatibility between control performance and markings and that the knob is secure to its shaft. Observe positive switch indexing and ease of minipulation of both controls.

4.6.4.17 Arm 10547473 operability.- This test consists of applying normal to Arm 10547473 the direct forces that equal the torque values in

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3.7.14. Actual values of direct force required for each examination shall be determined by the formula:

$$\text{Direct Force Applied (pounds)} = \frac{\text{Torque value in 3.7.14 (Pound inches)}}{\text{Arm Length Upon Which Force Acts (Inches)}}$$

Arm length upon which force acts shall be measured along Arm 10547473 toward the beamsplitter from an axis through the center of the bore in the yoke casting supporting Arm 19547473. Place Arm 10547473 in an operating position with the force gage (see 4.5.2.4) inserted - weighted end down - between the stop surfaces. Use a spring scale and apply sufficient force to cause the gage to fall clear. The measured value of slowly applied force that moves Arm 10547473 a distance in excess of 0.002 inch, which allows the gage to fall clear, shall be used to verify compliance with the applicable requirement in 3.7.14. Now use the spring scale and measure the force required to freely move Arm 10547473 from (not out of) its operating position to its stowed position and verify compliance with the applicable requirement in 3.7.14. Arm 10547473 shall then be placed in a stowed position. Again use the spring scale and measure the force required to cause the restraining linkage to fold permitting Arm 10547473 to move freely toward its operating position.

4.6.4.18 Range knob torque.- To perform this test, utilize a standard torque wrench with appropriate adapter. The range knob shall be rotated throughout the ranges in Column 1 of Table I in both cw and ccw directions. Observe that torque readings remain within the limits specified in 3.7.15, each time the knob is moved out of detent in both a cw and ccw direction.

NOTE: It is necessary to depress the detent plunger 10547323 to change from one scale to the other.

4.6.4.19 Sealing.- This test shall be performed prior to the humidity test 4.6.3 - when a humidity test is required (see 4.4) - utilizing test equipment conforming to 4.5. Housing 10547482 shall be internally pressurized with dry nitrogen gas to the pounds-per-square-inch gage pressure (psig) in 3.7.16 for not less than two hours. Throughout this period, no pressure drop greater than the limit specified in 3.7.16 shall be evident. Upon completion of this test, flush the housing with dry nitrogen gas through a regulator to prevent excessive internal pressure during flushing. Utilize a dew point tester and continue flushing until the gas bled from the housing exhibits a dew point at least as low as -25°F with an internal pressure of between 0.10 and 0.25 psig. Housing 10547482 shall then be sealed.

4.6.4.20 Interior atmosphere.- This test shall utilize the procedure in 4.6.4.19 except that the examination previously performed to detect excessive loss of pressure need not be repeated. This examination shall

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verify compliance with 3.7.17 and be performed subsequent to all tests and examinations of 4.6 except 4.6.4.21.

4.6.4.21 Cleanliness.- Subsequent to conducting all tests and examinations, optics of the projector shall be examined in accordance with MIL-O-13830 to verify compliance with 3.3.1b.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking of Reflex Sight: XM70E1 shall be in accordance with Packaging Data Sheet 10547304. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The reflex sight covered by this specification is primarily intended for use by the pilots of light observation helicopters (OH-6A) and OH-58A to obtain a high probability first round hit. It is intended to be installed on the aircraft centerline and be energized from the aircraft internal electrical system. A mechanical linkage connected to the gun drive moves the reflex sight in synchronism with the weapon axis in elevation and depression. The reflex sight has an azimuth adjustment for boresighting. The sight is optimized for use at a speed of 100 knots and an altitude of 50 meters and provided with range settings for 7.62mm and 40mm ammunition.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Provisions for first article testing (see 4.2).
c. Selection of applicable levels of preservation, packaging and packing (see 5.1).
d. Applicable packaging data sheet (see 5.1).
e. Applicable stock number.

Custodian:

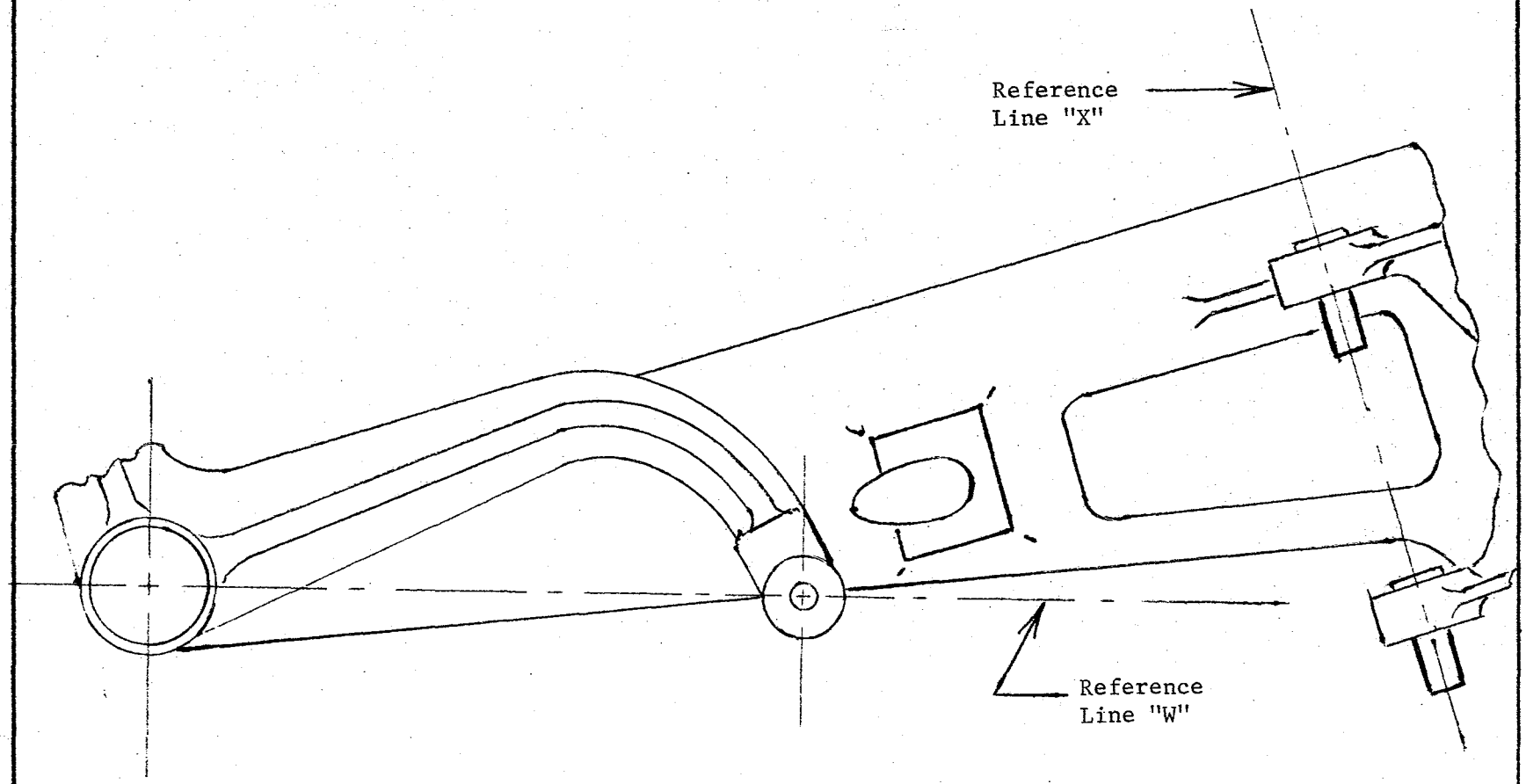
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Preparing activity:

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SUPPORT ASSEMBLY

FIGURE 1.

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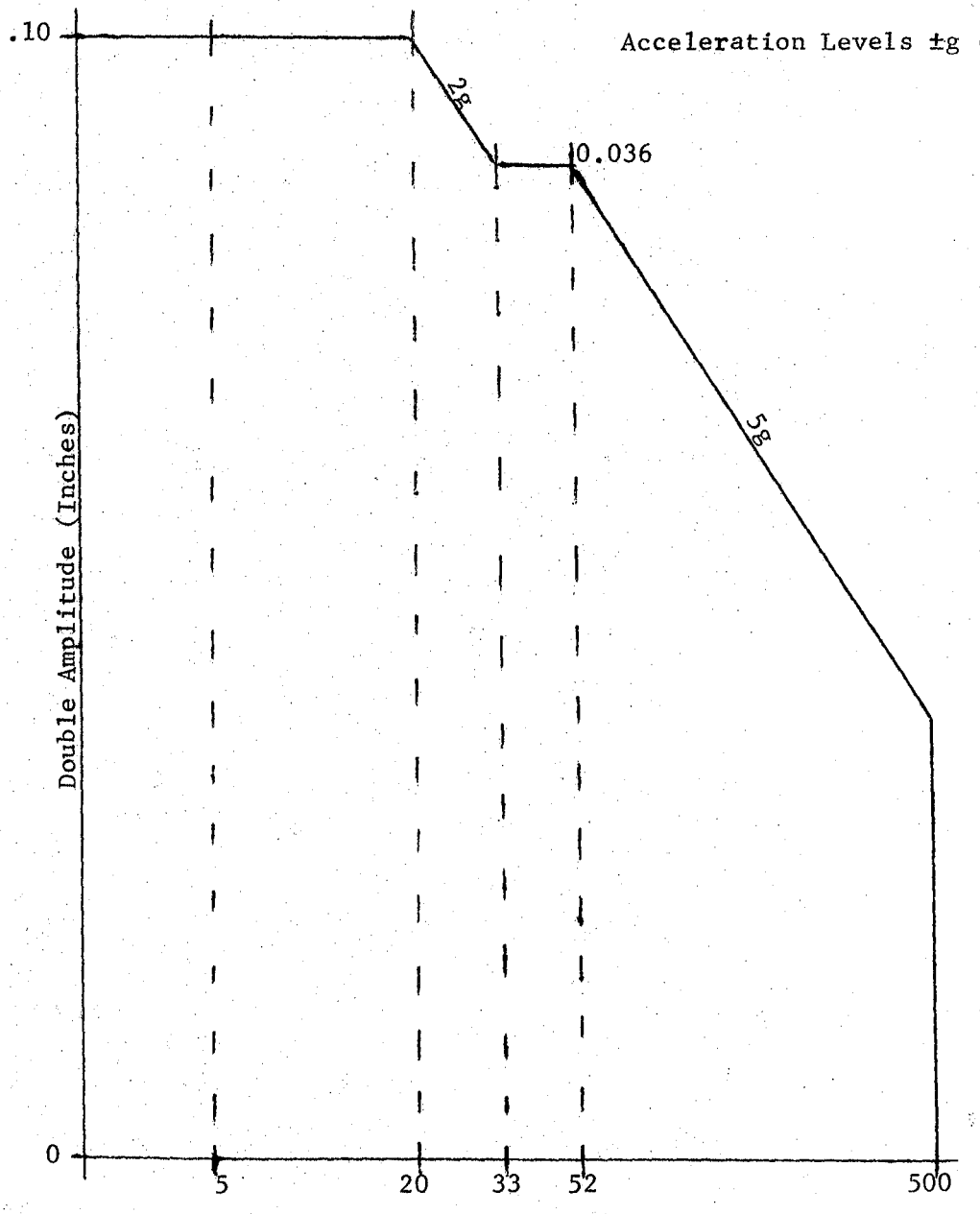


FIGURE 2. VIBRATION "A" CURVE

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